

MILOSEC[®] Capsule 20 mg (*Omeprazole* 20 mg)

Composition:

Each capsule contains Omeprazole Pellets equivalent to Omeprazole 20 mg

Product Description:

Opaque pink-reddish brown capsule No.2, imprinted "FrX FrX" in black color, containing off-white enteric coated pellet.

Pharmacodynamics:

Omeprazole reduces gastric acid secretion through a highly selective mechanism of action. It produces specific dose-dependent inhibition of the enzyme H^+ , K^+ -ATPase (the proton pump) in the parietal cell. As this action inhibits the final stage of gastric acid formation, there is an effective inhibition of both basal and stimulated acid secretion irrespective of the stimulus to acid formation. Omeprazole has no effect on acetylcholine or histamine receptors and no clinically significant pharmacodynamic effects have been observed other than those explained by the effect of omeprazole on acid secretion. The onset of action is rapid and reversible control of gastric secretion is achieved with once-daily dosing.

During treatment with antisecretory medicinal products, serum gastrin increase in response to the decreased acid secretion. Also CgA increases due to decreased gastric acidity. The increased CgA level may interfere with investigations for neuroendocrine tumors.

Available published evidence suggested that proton pump inhibitors should be discontinued between 5 days and 2 weeks prior to CgA measurements. This is to allow CgA levels that might be spuriously elevated following PPI treatment to return to reference range.

Pharmacokinetics:

Omeprazole is rapidly but variably absorbed after oral doses. Absorption is not affected by food.

Omeprazole is acid-labile and pharmacokinetics may vary between the various formulations developed to improve oral bioavailability. The absorption of omeprazole also appears to be dose-dependent; increasing the dosage above 40 mg has been reported to increase the plasma concentrations in a non-linear fashion because of saturable first-pass hepatic-metabolism. In addition, absorption is higher after long term use. Omeprazole is almost completely metabolized in the liver, primarily by the cytochrome P450 isoenzyme CYP2C19 to form hydroxyomeprazole and to a small extent by CYP3A to form omeprazole sulfone. The metabolites are inactive, and are excreted mostly in the urine and to a lesser extent in bile. The elimination half-life from plasma is reported to be about 0.5 to 3 hours. Omeprazole is about 95% bound to plasma-proteins.

Mode of Administration:

Oral route

Indications:

Treatment of duodenal ulcer, gastric ulcer and reflux oesophagitis and symptomatic gastroesophageal reflux disease. *Helicobacter pylori* eradication in peptic ulcer disease. Management of Zollinger-Ellison syndrome.

Dosage:

Duodenal Ulcer: Recommended Dosage: Milosec 20 mg once daily. Symptom relief is rapid and in most patients, healing occurs within 2 weeks. For those patients who may have fully healed after the initial course, healing usually occurs during a further 2-week treatment period. In patients with poorly responsive duodenal ulcer, Milosec 40 mg once daily has been used and healing is usually achieved within 8 weeks. For the prevention of relapse in patients with duodenal ulcer, the recommended dose is Milosec 20 mg once daily. If needed, the dose can be increased to 40 mg once daily. The effectiveness of Milosec is not affected by concomitant NSAID treatment, and the usual duration of treatment is recommended.

Gastric Ulcer: Recommended Dosage: Milosec 20 mg once daily. Symptom relief is rapid and in most patients, healing occurs within 4 weeks. For those patients who may not have fully healed after the initial course, healing usually occurs during a further 4-weeks treatment period. In patients with poorly responsive gastric ulcer, Milosec 40 mg once daily has been used and healing is usually achieved within 8 weeks. For the prevention of relapse in patients with poorly responsive gastric ulcer, the recommended dose is Milosec 20 mg once daily. If needed, the dose can be increased to Milosec 40 mg once daily. The effectiveness of Milosec is not affected by concomitant NSAID treatment and the usual duration of treatment is recommended.

***Helicobacter pylori* (Hp) Eradication Regimens in Peptic Ulcer Disease:**

Triple Therapy Regimens: Milosec 20 mg, amoxicillin 1 g and clarithromycin 500 mg, all twice a day for 1 week; or Milosec 20 mg, clarithromycin 250 mg and metronidazole 400 mg (or tinidazole 500 mg), all twice a day for 1 week; or Milosec 40 mg once daily with amoxicillin 500 mg and metronidazole 400 mg both 3 times a day for 1 week.

Dual Therapy Regimens: Milosec 40-80 mg daily with amoxicillin 1.5 g daily in divided doses for 2 weeks; or Milosec 40 mg once daily and clarithromycin 500 mg 3 times a day for 2 weeks. To ensure healing in patients with active ulcer diseases, see further dosage recommendations for duodenal and gastric ulcer.

In each regimen, if the patient is still Hp positive, therapy may be repeated.

Reflux Esophagitis: Recommended Dosage: Milosec 20 mg once daily. Symptom relief is rapid and in most patients, healing occurs within 4 weeks. For those patients who may not have fully healed after the initial course, healing usually occurs during a further 4-week treatment period. In patients with severe reflux esophagitis, Milosec 40 mg once daily has been used and healing is usually achieved within 8 weeks. For the long-term management of patients with healed reflux esophagitis, the recommended dose is Milosec 20 mg once daily. If needed, the dose can be increased to 40 mg once daily.

Symptomatic Gastroesophageal Reflux Disease: Recommended Dosage: Milosec 20 mg daily. Symptom relief is rapid. Patients may respond adequately to 20 mg daily, and therefore individual dose adjustment should be considered. If symptom control has not been achieved after 4-week treatment with Milosec 20 mg daily, further investigation is recommended.

Zollinger-Ellison Syndrome: Recommended Initial Dosage: Milosec 60 mg daily. The dosage should be adjusted individually and treatment continued as long as clinically indicated. All patients with severe disease and inadequate response to other therapies have been effectively controlled and >90% of the patients maintained on doses of Milosec 20-120 mg daily. With doses >80 mg daily, the dose should be divided and given twice daily.

Impaired Renal and Liver Function: Dose adjustment in patients with impaired renal or liver function is not required.

Children: There is no experience with Milosec in children.

Elderly: No dose adjustment is necessary in the elderly.

Contraindications:

Known hypersensitivity to omeprazole.

Precautions:

- Should not be used in children
- Should not be given during pregnancy and lactation unless its use is considered essential.
- When gastric ulcer is suspected, the possibility of malignancy should be excluded as treatment may alleviate symptoms and delay diagnosis

Regular Surveillance

Patients on proton pump inhibitor treatment (particularly those treated for long term) should be kept under regular surveillance.

Subacute Cutaneous Lupus Erythematosus (SCLE)

Proton pump inhibitors are associated with very infrequent cases of subacute cutaneous lupus erythematosus (SCLE). If lesions occur, especially in sun-exposed areas of the skin, and if accompanied by arthralgia, the patient should seek medical help promptly and the health care professional should consider stopping Milosec. SCLE after previous treatment with proton pump inhibitor may increase the risk of SCLE with other proton pump inhibitors.

Hypomagnesaemia

Severe hypomagnesaemia has been reported in patients treated with PPI like Milosec for at least three months, and in most cases for a year. Serious manifestations of hypomagnesaemia such as fatigue, tetany, delirium, convulsions, dizziness and ventricular arrhythmia can occur but they may begin insidiously and be overlooked. In most affected patients, hypomagnesaemia improved after magnesium replacement and discontinuation of the PPI.

For patients expected to be on prolonged treatment or who take PPI with digoxin or drugs that may cause hypomagnesaemia (e.g., diuretics), health care professionals should consider measuring magnesium levels before starting PPI treatment and periodically during treatment.

Fracture

Proton pump inhibitors, especially if used in high doses and over long durations (>1 year), may modestly increase the risk of hip, wrist and spine fracture, predominantly in the elderly or in presence of other recognized risk factors. Observational studies suggest that proton pump inhibitors may increase the overall risk of fracture by 10-40%. Some of this increase may be due to other risk factors. Patients at risk of osteoporosis should receive care according to current clinical guidelines and they should have an adequate intake of vitamin D and calcium.

Clostridium Difficile Diarrhea

Published observational studies suggest that PPI therapy may be associated with an increased risk of Clostridium difficile associated diarrhea, especially in hospitalized patients. This diagnosis should be considered for diarrhea that does not improve. Patients should use the lowest dose and shortest duration of PPI therapy appropriate to the condition being treated.

Vitamin B12 Deficiency

Daily treatment with any acid-suppressing medications over a long period of time (e.g., longer than 3 years) may lead to malabsorption of cyanocobalamin (vitamin B12) caused by hypo- or achlorhydria. Rare reports of cyanocobalamin deficiency occurring with acid-suppressing therapy have been reported in the literature. This diagnosis should be considered if clinical symptoms consistent with cyanocobalamin deficiency are observed.

Increased Chromograin A (CgA) level may interfere with investigations for neuroendocrine tumors. If the patient(s) are due to have a test on Chromogranin A level, Milosec treatment should be stopped for at least 5 days before CgA measurements to avoid this interference (see Pharmacodynamic section). If CgA and gastrin level have not returned to reference range after initial measurement, measurements should be repeated 14 days after cessation of proton pump inhibitor treatment.

Interactions with Other Medicaments:

Milosec can prolong the elimination of diazepam, warfarin and phenytoin, drugs that are metabolised by oxidation in the liver. Monitoring of patients also receiving warfarin or phenytoin is recommended and a reduction of dose of warfarin and phenytoin may be necessary. No interaction with propranolol or theophylline has been found, but there may be interactions with other drugs also metabolised via the cytochrome P-450 enzyme system. No interactions with concomitantly administered antacids have been found.

Pregnancy and Lactation:

Milosec should not be given during pregnancy and lactation unless its use is considered essential. Animal studies have not shown evidence of any hazard from the administration of Milosec during pregnancy and lactation and there is no evidence of foetal toxicity or teratogenic effect.

Side Effects:

Skin:

Skin rash, urticaria and pruritus have been reported, usually resolving after discontinuation of treatment.

Central and peripheral nervous system:

Headache is a common side effect. Dizziness, paraesthesia, somnolence, insomnia and vertigo are rare. In isolated cases, reversible mental confusion, agitation, depression and hallucination predominantly in severely ill patients have occurred. Increased sweating, peripheral oedema and blurred vision have been reported in isolated cases.

Gastrointestinal:

Diarrhoea, constipation, nausea, vomiting, flatulence and abdominal pain.

Hepatic:

Rarely increased in liver enzymes. In isolated cases, encephalopathy in patients with pre-existing severe liver disease; hepatitis with or without jaundice and hepatic failure have been reported.

Endocrine:

In isolated cases, gynaecomastia has been reported

Haematological:

In isolated cases, leukopenia, thrombocytopenia, agranulocytosis and pancytopenia have been reported.

Musculoskeletal:

In isolated cases, arthralgia, muscular weakness and myalgia have been reported.

Subacute Cutaneous Lupus Erythematosus (SCLE):

Skin and subcutaneous tissue disorders

Frequency 'not known': Subacute cutaneous lupus erythematosus

Interstitial Nephritis:

Renal and urinary disorders: Interstitial nephritis

Hypomagnesaemia:

Metabolism and nutritional disorders

Frequency 'not known': hypomagnesaemia.

Fracture:

Musculoskeletal disorders

Frequency 'uncommon': Fracture of the hip, wrist or spine.

Clostridium Difficile Diarrhea:

Infections & infestations: Clostridium difficile associated diarrhea.

Fundic Gland Polyps (Benign):

Gastrointestinal disorders

Frequency 'common': Fundic gland polyps (benign)

Vitamin B12 Deficiency:

Metabolic/Nutritional: Vitamin B12 deficiency

Gastrointestinal disorders:

Microscopic colitis: Frequency 'not known'

Symptoms and Treatment of Overdose:

There is no information available on the effects of overdosage in man and specific recommendations for treatment cannot be given. Single oral doses of up to 400 mg have been well tolerated.

Signs and Symptoms: The following effects have been selected on the basis of their potential Clinical significance, such as: blurred vision; confusion; diaphoresis (increased sweating); drowsiness; dryness of mouth; flushing; headache; nausea; tachycardia (fast or irregular heartbeat)

Treatment of overdose: Since there is no specific antidote for overdose with omeprazole, treatment should be symptomatic and supportive. Due to extensive protein binding, omeprazole is not readily dialyzable. Patients in whom intentional overdose is confirmed or suspected should be referred for psychiatric consultation.

Packaging Available

Blister of 2x7 capsules / carton

Store below 30°C and protect from light.

Keep medicines out of reach of children.

Jauhi daripada kanak-kanak.

Shelf life

3 years

Manufactured for:

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